NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

MIL-PRF-39012/89B
3 October 1986
SUPERSEDING
MIL-C-39012/89A
28 July 1983

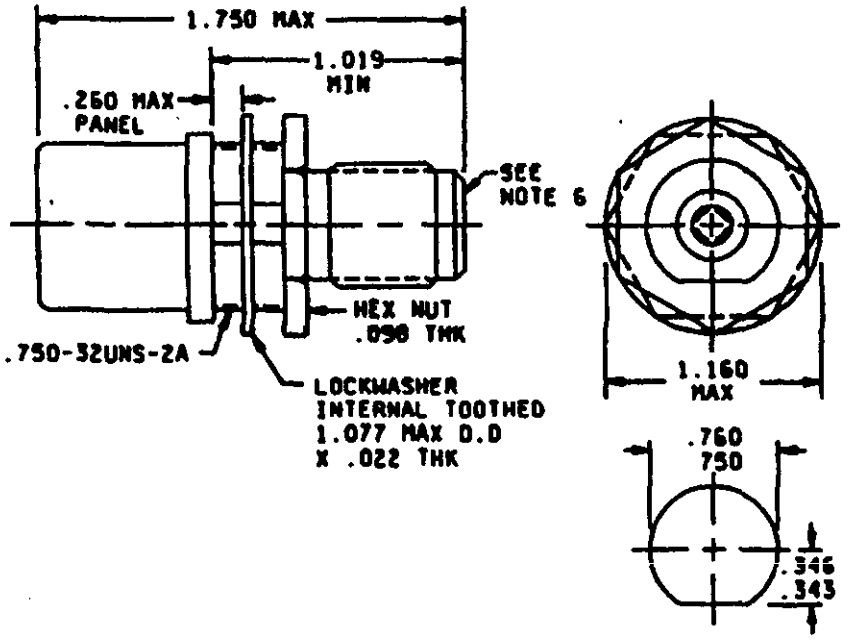
PERFORMANCE SPECIFICATION

CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES QSC, (CABLED, SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-PRF-39012

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE III



Inches	mm
022	0 56
098	2 49
260	6 60
343	8 71
346	8 79
750	19 05
760	19 30
1 019	25 88
1 077	27 38
1 169	29 46
1 750	44 45

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Dimension 1 750 (44 45 mm) defines the overall length when assembled to a connector
4. Dimension 1 160 (29 46 mm) defines the overall diameter of the connector
5. Wrench flats to accommodate standard wrench opening in accordance with FED-STD-H28 appendix 10
6. Series QSC socket contact interface per MIL-STD-348

FIGURE 1 General configuration

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

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TABLE I. Dash number and usage cross-reference.

Dash number 1/	Applicable cable no. ≅
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 4/	
0001 5/	M17/28-RG058 M17/155-00001ø M17/60-RG142ø M17/158-00001ø M17/84-RG223* M17/167-00001ø
0002 5/	M17/29-RG059*Δ M17/30-RG062Δ M17/90-RG071Δ M17/110-RG302øΔ M17/97-RG210Δ
0003 5/	M17/54-RG122* M17/157-00001ø
0004 5/	M17/111-RG303ø* M17/170-00001ø
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520/5-01 BASIC CRIMP TOOL) 2/	
0010 3/	M17/28-RG058*~ M17/155-00001~ø M17/111-RG303ø~ M17/170-00001~ø
0011 3/	M17/60-142ø~ M17/158-00001~ø M17/84-RG223*~ M17/167-00001~ø
0012 3/	M17/29-RG059*Δ> M17/30-RG062Δ> M17/110-RG302øΔ> M17/97-RG210Δ>
0013 3/	M17/54-RG122*& M17/157-00001&ø
0014 3/	M17/90-RG071*Δ>

1/ Typical mating connectors for each dash number are those connectors of MIL-C-39012/65 which are of the same category and utilize the cable.

2/ Category C connectors are assembled by means of the applicable crimping tool per MIL-C-22520 to the specified cable stripped in accordance with figure 2.

3/ These connectors have captivated center contacts.

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4/ Inactive for new design. Not for Air Force or Navy use.

5/ These connectors shall be furnished with noncaptive center contacts.

≡ The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the ±200°C thermal shock tests.

Δ These are not 50 ohm cables, therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.

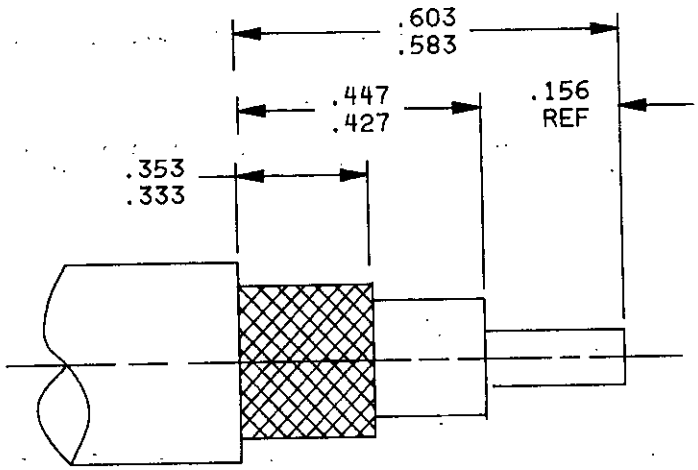
& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

~ M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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Inches	mm
.156	3.96
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.583	14.81
.603	15.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Cable stripping dimensions for category C connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

500 volts rms maximum working voltage.

125 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 2 inch-pounds maximum.

Coupling proof torque: 20 inch-pounds minimum.

Inspection conditions: 4 to 6 inch-pounds.

Mating characteristics:

See MIL-STD-348.

Center contact (socket):

Oversize test pin - .057 diameter minimum (nonclosed entry contacts only).

Insertion depth - .125 minimum.

Number of insertions - One.

Insertion force test - Steel test pin diameter .054 minimum.

Test pin finish - 16 microinches.

Insertion force - 2 pounds maximum.

Withdrawal force test: Steel test pin diameter .052 maximum.

Withdrawal force - 2 ounces minimum.

Test pin finish - 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: 6 pounds minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR):

From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower. 1.30 maximum.

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Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.003 + .002 F (F in GHz).

Item 16 - VSWR shall be less than 1.008 + .002 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.012 + .004 F (F in GHz).

Group B inspection - VSWR shall be less than 1.024 +.007 F (F in GHz).

Group C inspection - VSWR shall not exceed 1.08.

Connector durability:

500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact:	1.5	2.0
Outer contact:	.2	Not applicable
Braid to body:	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,000 volts rms at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B.

Shock: Method 213 of MIL-STD-202, test condition I.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and II).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude - 70,000 feet.

Voltage - 375 volts.

RF high potential withstanding voltage:

Voltage and frequency - 1,000 volts rms at 5 MHz.

Leakage current - Not applicable.

Cable retention force:

Non-crimp assemblies: 40 pounds minimum.

Crimp assemblies:

50 pounds minimum for cables .155-.189 OD.
60 pounds minimum for cables .190-.229 OD.
75 pounds minimum for cables .230-.249 OD.
90 pounds minimum for cables .250-and larger.

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Coupling mechanism retention force: Not applicable.

RF leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .2 dB maximum at 3 GHz.

Qualification not required.

Part number: M39012/89 (dash number from table I or "B" number from table II).

TABLE II. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY

M39012/89B 1/ 2/ 3/	Applicable cable
0005	MI7728-RG058* M17/155-00001ø
0006	MI7784-RGZ23* M17/167-00001ø
0007	MI7729-RG059*Δ M17/30-RG062Δ M17/110-RG302@Δ M17/97-RG210Δ
0008	MI7754-RGI22* M17/157-00001ø
0009	MI7790-RG07I*Δ
0015	MI77III-RG303@* M17/170-00001ø
0016	MI7760-RGI42@* M17/158-00001ø

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2/ These connectors shall be furnished with noncaptive center contacts.

3/ Inactive for new design.

≅ The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C thermal shock tests.

Δ These are not 50 ohm cables, therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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TABLE III. Cross-reference of part numbers.

Part number 1/ M39012/89B	Superseded part number M39012/89-
0005	0005
0006	0006
0007	0007
0008	0008
0009	0009
0015	0015
0016	0016

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, EA
Navy - SH
Air Force - 11, 17, 80, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3518-35)

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